

Homework 01 Solutions

Author: Dustin Miao (dustinmi)
Course: 10-783 A
Due by: September 22, 2026

Task 1.1

We have that X is a zero-centered Gaussian since e^{-t} is a deterministic function of time. Then, by Ito isometry,

$$\text{Var}(X) = \mathbf{E} \left[\left(\int_0^2 e^{-t} dW_t \right)^2 \right] = \int_0^2 e^{-2t} dt = \frac{1 - e^{-4}}{4}.$$

Thus, $X \sim \mathcal{N}(0, (1 - e^{-4})/4)$.

Task 1.2

Let $f(t)$ be the piecewise function

$$f(t) = \begin{cases} 1 & 0 \leq t \leq 1/2 \\ 2 & 1/2 \leq t \leq 1 \end{cases}.$$

Because $f(t)$ is time-dependent, X is a zero-centered Gaussian with variance

$$\text{Var}(X) = \mathbf{E} \left[\left(\int_0^1 f(t) dW_t \right)^2 \right] = \int_0^1 f(t)^2 dt = 2.5.$$

Thus, $X \sim \mathcal{N}(0, 2.5)$.

Task 1.3

We have that

$$\begin{aligned} \text{Var}(X) &= \mathbf{E} \left[\left(\int_0^1 t dW_t \right)^2 \right] = \int_0^1 t^2 dW_t = 1/3 \\ \text{Var}(Y) &= \mathbf{E} \left[\left(\int_0^1 (1-t) dW_t \right)^2 \right] = \int_0^1 (1-t)^2 dW_t = 1/3 \\ \text{Cov}(X, Y) &= \int_0^1 t(1-t) dt = 1/6. \end{aligned}$$

Thus, X and Y are not independent because they have a non-zero covariance.

Task 1.4

Task 1.10

(a) True. We have that

$$\int_0^1 f(t) dW_t = \lim_{|\Pi| \rightarrow 0} \sum_{i=1}^{n-1} f(t_i)(W_{t_{i+1}} - W_{t_i}),$$

where each interval $W_{t_{i+1}} - W_{t_i} \sim \mathcal{N}(0, \Delta t_i)$ and $W_{t_{i+1}} - W_{t_i} \perp W_{t_{j+1}} - W_{t_j}$ for $i \neq j$. Thus, for any partition Π , $\int_0^1 f(t) dW_t$ is a weighted sum of independent Gaussians, so it must be normally distributed.

Task 2

(1)

t	E[W_n(t)]	theory=0	Var[W_n(t)]	theory=t
0.100	-0.00473	0.000	0.10167	0.100
0.250	-0.00961	0.000	0.25102	0.250
0.500	-0.01644	0.000	0.50755	0.500
0.750	-0.02026	0.000	0.77095	0.750
1.000	-0.02012	0.000	1.02874	1.000

(2) See Figure 1.

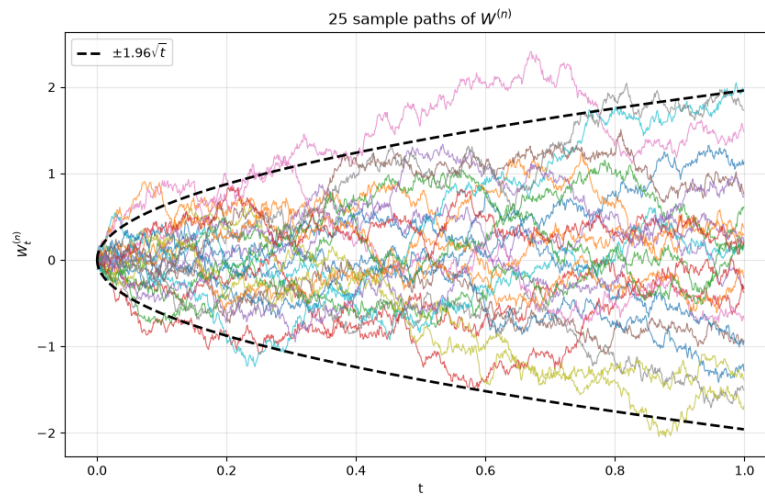


Figure 1: 25 generated Wiener processes with 95% envelope

(3) See Figure 2.

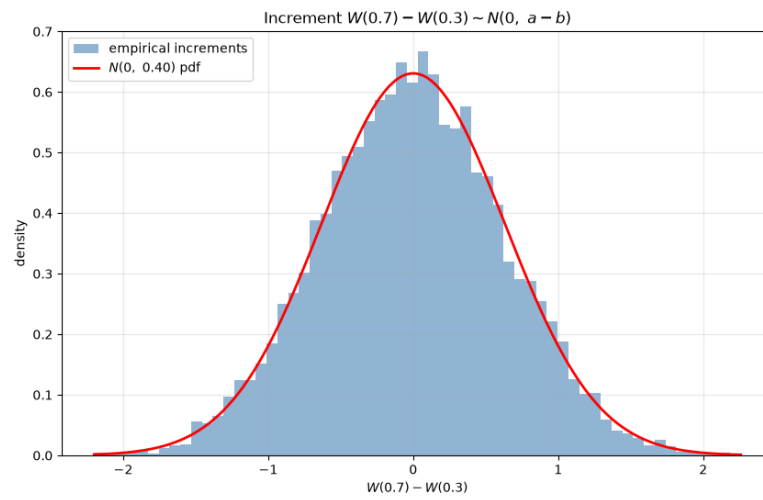


Figure 2: Empirical distribution of Wiener process increments

Collaboration Questions

- (1) (a) Did you receive any help whatsoever from anyone in solving this assignment? No.
 (b) If you answered 'yes', give full details. N/A.
- (2) (a) Did you give any help whatsoever to anyone in solving this assignment? No.
 (b) If you answered 'yes', give full details. N/A.
- (3) (a) Did you find or come across code that implements any part of this assignment? No.
 (b) If you answered 'yes', give full details. N/A.